Employee table
Employee(<u>EmpID</u>, Email, PhoneNum, FName, MiddleI, LName, SectionNum)

FD1: EmpID -> Fname, MiddleI, LName

The employee table is in 1NF, since it has only atomic values, so no multivalued or composite attributes
It is also in 2NF, since the only functional dependency is from a complete key to non-prime values
It is also in 3NF, since there are no transitive dependencies

User Table
User(<u>UserID</u>, Email, PhoneNum, FName, MiddleI, LName)

FD1: UserID -> Fname, MiddleI, LName

The User table is in 1NF due to having only atomic values
It is also in 2NF, since the only functional dependency is from a complete key to non-prime values
It is also in 3NF, since there are no transitive dependencies

Book Table
Book(<u>BookID</u>, Author, Title, Publisher, PubDate, Genre, IsRented, SectionNum)

FD1: BookID -> Author, Title, Publisher, PubDate, Genre, IsRented, SectionNum

The Book Table is in 1NF due to having only atomic values
It is also in 2NF since all non-prime attributes are dependent on the complete key
It is also in 3NF since there are no transitive dependencies

Section Table
Section(<u>SecNum</u>, Kids/Adult)

FD1: SecNum -> Kids/Adult

The SectionTable is in 1NF due to having only atomic values
It is also in 2NF since all non-prime attributes are dependent on the complete key
It is also in 3NF since there are no transitive dependencies

Rent/Return Table
Rent(<u>UserID, BookID</u>, Rent_Date, Return_Date)

FD1: UserID, BookID -> Rent_Date
FD2: Rent_Date -> Return_Date

The Rent Table is NOT in 3NF due to having a transitive FD, thus will need to be normalized.

In this table, our X -> Y, Y -> Z ⇒ (UserID, BookID) -> Rent_Date, Rent_Date -> Return_Date

Y^+ = { Rent_Date, Return_Date)
Let $R_1 = Y^+$
Let $R_2 = (R - Y^+) \cup Y$

R_1(<u>Rent_Date</u>, Return_Date)
R_2(<u>UserID, BookID</u>, Rent_Date)

FD's have to be maintained, thus R_1 has
FD: Rent_Date -> Return_Date

And R_2 has
FD: (UserID, BookID) -> Rent_Date

R_1 and R_2 are both in 1NF due to having no atomic values, and are also in 2NF due to all none-prime values being determined by a complete key, not a partial key. They are both now in 3NF due to having no transitive dependencies